

1. System Overview

The **Flutter Hybrid TikTok–Instagram Application** is a mobile application that combines features from both TikTok and Instagram. The system allows users to create accounts, upload photos or videos, view posts in a feed, and interact with other users through likes and comments.

The application uses **Flutter as the frontend framework**, while **Cloudinary serves as the media storage system** for images and videos. The app follows a **client-cloud architecture**, where the mobile app interacts with cloud services to upload, retrieve, and display media content.

2. System Architecture

The system uses a **Hybrid Mobile Architecture** composed of three main layers:

1. Presentation Layer (Flutter App)

This layer manages the **user interface and user interaction**.

Responsibilities:

- Display feeds
- Upload media
- Show user profiles
- Handle navigation

Technologies used:

- Flutter
- Provider (state management)
- Go Router (navigation)

2. Application Layer (App Logic)

This layer processes the application logic such as:

- Media uploading
- State management
- Content processing

Main services:

Service	Function
Cloudinary Service	Upload images/videos
Post Provider	Manage post data
Media Picker	Select images/videos

3. Cloud Layer (External Services)

This layer stores and delivers application data.

Services used:

Service	Purpose
Cloudinary	Store images and videos
CDN Delivery	Fast media loading
Secure URL	Access uploaded files

Cloudinary handles:

- media storage
- media optimization
- CDN distribution

3. Core System Modules

3.1 User Interface Module

This module handles the **visual components of the app**.

Main screens include:

- Home Feed (TikTok-style scrolling)
- Upload Screen
- Profile Page
- Video Viewer

Flutter widgets are used to create responsive UI components.

3.2 Media Upload Module

The system allows users to upload images or videos using the device's gallery or camera.

Process:

1. User selects media
2. Media is uploaded to Cloudinary
3. Cloudinary returns a secure URL
4. The URL is stored and displayed in the app

Package used:

- image_picker
- clouinary_public

3.3 Feed Display Module

The feed shows uploaded content similar to TikTok and Instagram.

Features:

- Vertical scrolling video feed
- Image post display
- Cached media loading

Packages used:

- video_player

- `cached_network_image`

3.4 State Management Module

The app uses **Provider** to manage application data.

This module handles:

- Post data
- Upload state
- UI updates

Example state data:

- list of posts
- uploaded media URLs

- user interactions

4. Data Flow Overview

The system follows this general flow:

1. The user opens the application.
2. The user selects media from the device.
3. The media file is uploaded to Cloudinary.
4. Cloudinary stores the file and returns a secure URL.
5. The application displays the uploaded media in the feed.

Simplified flow:

User → Pick Media → Upload to Cloudinary → Receive URL → Display Post

5. System Components

Frontend

Flutter Mobile Application

Packages used:

- provider
- go_router
- image_picker
- video_player
- cached_network_image

Cloud Services

Cloudinary

Used for:

- image hosting
- video hosting
- CDN delivery
- media optimization

6. System Advantages

The architecture provides several benefits:

Scalability

Cloudinary handles large amounts of media storage.

Performance

CDN delivery ensures fast media loading.

Simplicity

The system does not require a complex backend server.

Cross-Platform Support

Flutter allows the app to run on:

- Android

- iOS
- Web (hybrid capability)

7. Future Enhancements

Possible improvements include:

- User authentication system
- Comment and like functionality
- Real-time notifications
- Video editing features
- Cloud database integration

8. Summary

The **Flutter Hybrid TikTok–Instagram Architecture** uses a **client-cloud model** where the Flutter application manages the user interface and user interactions while Cloudinary handles media storage and delivery.

This architecture ensures that the application remains **lightweight, scalable, and efficient** while providing a smooth media-sharing experience similar to TikTok and Instagram.

If you want, I can also give you a **simple architecture diagram for this system** that looks like a **professional software architecture (Flutter → Cloudinary → Media CDN)** which is perfect for documentation or presentations.